Response to the Exp2/Exp3 Results Audit: Recommended Next Steps

Working note — CAROM series

July 2026

1 On the audit itself

The audit is excellent work. The observed/supplied/inferred/not-established taxonomy, the hash manifest, the all-negative-baseline correction on edge accuracy, and the paired-versus-unpaired catch on M2 are exactly the epistemics this program needs. Two of its findings are flaws in my apparatus and should be owned as such: the shared-RNG evaluation (evaluation calls consuming the training sample stream, no fixed corpora) and the unpaired M2 are design shortcuts from the original sandbox harness — which means the earlier sandbox interference numbers (0.772 $\to$ 0.776; the 0.65 "recovery") carry the same invalidity the audit assigns to the GPU run's. The `sq` paraphrase discrepancy (0.438 sandbox vs. 0.275 retrieved) is the multi-seed warning materializing on schedule.

2 The two signals that decide the program, and their cheap discriminating experiments

The audit's P0–P2 priorities are sound, but two results deserve resolution *before* any new training run, because both have inexpensive discriminating experiments and both outcomes reshape what the next run should be.

2.1 The tau decline (0.990 $\to$ 0.492): drift or metric failure

This is the most important number in the document. Itinerary tau falling while endpoint accuracy rises is either *mechanistic drift* — the model abandoning itinerant execution for distributed operator mixing, which is precisely the phase-smearing failure mode predicted in the itinerant note, now possibly observed (the architecture *permits* a non-itinerant solution and the task gradient found it) — or *metric failure* (early near-perfect tau rewarding ordered subsets without coverage). Two steps distinguish them:

(1) **Retroactive re-measurement.** The checkpoints exist. Re-evaluate every saved checkpoint on one frozen corpus with repaired itinerary metrics (station coverage, transition precision/recall, exact-order rate, dwell and revisitation statistics). This converts the anecdotal decline into a true mechanistic-drift curve — or dissolves it.

(2) **Trajectory intervention (decisive).** At evaluation, clamp the control state: force the correct itinerary, force a shuffled one, force a single smeared mixture, and compare endpoint accuracy against the natural dynamics. If accuracy is indifferent to forced wrong orderings, the channel is decorative and the causal-heteroclinic-execution claim is dead for this checkpoint. If accuracy collapses under shuffled itineraries and recovers under forced-correct ones, the channel is load-bearing and the drift is partial. This single intervention does more causal work than the full ablation grid, costs one afternoon on existing checkpoints, and should run before any retraining.

If drift is confirmed, the fixes are architectural rather than merely regularizing: sharpen the control–workspace coupling so updates require near-vertex control states (or straight-through top-1 mode gating), making the distributed shortcut structurally unavailable instead of penalized.

2.2 The L=5 failure (0.306): an unexamined time-budget confound

$S = 72$ integration steps was tuned for ≤ 4 phases at roughly 12–15 steps of dwell each; five phases need 60–75, so the trajectory may simply run out of clock — with the leak term degrading workspace content throughout. Before concluding that compilation fails to length-generalize, evaluate the *existing* L=5 checkpoint with S raised to 100–120 at test time. If accuracy jumps, the negative result is a budget artifact and the honest headline weakens to "extrapolation bounded by integration budget" — far more fixable. If it does not jump, the failure is real and the edge-scorer/compiler pathway is the suspect. Ten minutes of GPU either way.

3 Then, in order

(1) **Instrument repairs as a v2 harness, not patches.** Frozen train/validation/test corpora with independent RNG streams and example IDs; paired M2 with routing-versus-operator error decomposition; AUROC/AUPRC, positive recall, and exact-graph accuracy for edges; itinerary metrics validated on constructed positive and negative trajectories. The intervention tooling of §2.1 should be a permanent harness feature: "clamp the itinerary" is the reusable causal probe for everything downstream.

(2) **A reduced ablation grid.** Five seeds × six conditions is thirty runs; three conditions carry most of the causal information — task-only (edges=0), oracle-graph, shuffled-graph — bracketing the compiler's contribution from below, above, and adversarially. Add null-graph and the scheduled executor only if those three leave ambiguity.

(3) **Gate further TinyLM elaboration on completing the GPT-2 arm.** Three of the audit's not-established rows — semantic routing, language-compiled graphs, paraphrase transfer — bottleneck on the same variable, base semantic quality, which every sandbox session has isolated and no completed run has tested. The started-but-unfinished GPT-2 arm is the highest-information single run available. For Exp3, pair it with position-diverse anchors (current anchors are computed at first position and evaluation always invokes there; middle- and final-position invocation will likely degrade for the anchor-conditioning reason, not only the untested-position reason) and multi-skill sequential registration under

the paired M2, where the routing-interference law either compounds or is tamed by the KL no-regression penalty.

4 Standing

The audit's narrower defensible statement — frozen text representations support supervised dependency prediction and anchor-based registration of newly fitted operators, without yet reliable length or paraphrase generalization, and with causal execution and non-interference unvalidated — is one to co-sign as the program's current truth. What this note adds: the two questions named as unvalidated are not equally expensive to resolve. One is an afternoon of interventions on existing checkpoints; the other is a ten-minute budget check. That is where this week's effort belongs, before any new training run.